



COMMON NETWORK ISSUES

Networking – Theory & Practical Assignment

Topic: Common Hardware and Software Issues

Practical: Identifying Faulty Devices in a Network Simulator

Task: Creating a Network Troubleshooting Checklist

◆ TASK 1

📌 Question:

List 5 common hardware issues that can cause network problems in a typical office setup, and for each, describe one symptom that helps identify it.

✅ Answer:

Hardware problems are one of the most common causes of network failures. The following are five common hardware issues found in an office network:

No.	🔧 Hardware Issue	⚠️ Symptom
1	Damaged Ethernet Cable	The computer shows “ Network Cable Unplugged ” or has no network connection.
2	Faulty Network Card (NIC)	The computer cannot connect to the network even when the Ethernet cable is working.
3	Switch Failure	Several computers connected to the switch lose their network connection.
4	Router Failure	Users cannot access the internet or communicate with other networks.
5	Loose Cable Connection	The network connection frequently disconnects or the port/link light does not turn on.

💡 Conclusion:

Physical components such as **cables, switches, routers, and network cards** should be checked first because a simple hardware problem can stop network communication.

◆ TASK 2

📌 Question:

Open Cisco Packet Tracer or any network simulator, set up a small network with 3 PCs, 1 switch, and 1 router, and intentionally disconnect one device. Use the simulator's troubleshooting tools to identify which device is faulty and document the steps you took.

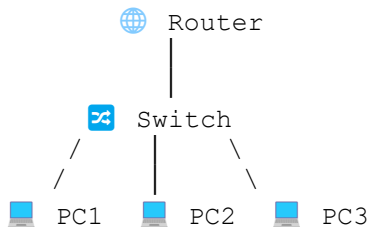
✅ Answer:

💻 Network Setup

The following devices were used:

-  PC1
-  PC2
-  PC3
-  1 Switch
-  1 Router

🏠 Network Topology



🔧 Troubleshooting Steps

Step 1: Opened Cisco Packet Tracer.

Step 2: Added 3 PCs, 1 switch, and 1 router to the workspace.

Step 3: Connected the devices using Ethernet cables.

Step 4: Configured the following IP addresses:

Device	IP Address
 Router	192.168.1.1
 PC1	192.168.1.10
 PC2	192.168.1.11
 PC3	192.168.1.12

Step 5: Tested connectivity using the `ping` command.

For example, from PC1:

```
ping 192.168.1.11
```

The ping was successful, which confirmed that the devices were communicating.

Step 6: Intentionally disconnected the Ethernet cable between **PC2 and the Switch**.

Step 7: Tested the connection from PC2 again.

```
ping 192.168.1.1
```

The ping failed.

Step 8: Checked the network topology and noticed that the connection between **PC2 and the Switch** was disconnected.

Step 9: Reconnected the Ethernet cable.

Step 10: Tested the connection again using `ping`.

The ping was successful after reconnecting the cable. 

Fault Identified:

Fault: Disconnected Ethernet cable between **PC2 and the Switch**.

Result:

Test	Before Fix 	After Fix 
PC2 → Router Ping	Failed	Successful
PC2 → Network	No Connection	Connected
Cable Status	Disconnected	Connected

Conclusion:

The faulty connection was identified by **checking the topology, observing the link status, and testing connectivity with ping**. Reconnecting the cable solved the problem.

TASK 3

Question:

Create a troubleshooting checklist for diagnosing slow internet issues in a home WiFi network, including at least 8 steps covering both hardware and software checks.

Hint: Think about checking cables, router status lights, device settings, and running ping or speed tests.

✓ **Answer:**

Home WiFi Slow Internet Troubleshooting Checklist

When internet speed becomes slow, follow these steps in order:

✓ **1. Check the Router Power**

Make sure the router is switched on and the **Power LED** is working properly.

✓ **2. Check Router Status Lights**

Check the **Internet/WAN, WiFi, and LAN lights**. An unusual or red light may indicate a problem.

✓ **3. Check Network Cables**

Make sure the **power and Ethernet/WAN cables** are properly connected and are not damaged.

✓ **4. Restart the Router**

Turn the router off, wait for a few seconds, and turn it back on. This can resolve temporary network problems.

✓ **5. Check WiFi Signal Strength**

Check the WiFi signal on the laptop or phone.

 If the signal is weak, move closer to the router and test again.

✓ **6. Check Connected Devices**

Check how many devices are connected to the WiFi. Too many active devices can reduce the available bandwidth.

✓ **7. Run an Internet Speed Test**

Run a speed test and compare the result with the speed provided by the ISP.

This helps determine whether the issue is with the WiFi network or the internet service.

✅ 8. Test the Connection Using Ping

Open **Command Prompt** and run:

```
ping 8.8.8.8
```

Check for:

- High response time
- Request timeouts
- Packet loss

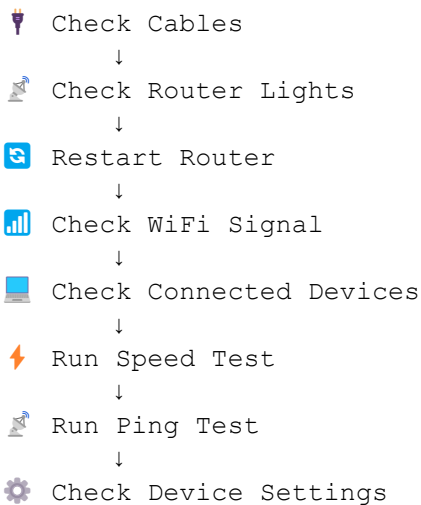
✅ 9. Check WiFi Settings

Make sure WiFi is enabled and the device is connected to the **correct WiFi network**.

✅ 10. Check Software and Drivers

Check for Windows updates and update the **WiFi/network adapter driver** if necessary.

Quick Troubleshooting Flow



💡 Conclusion:

Slow internet can be caused by **hardware problems, weak WiFi signals, too many connected devices, software issues, incorrect settings, or ISP problems**. Following a systematic checklist makes troubleshooting easier.

◆ TASK 4

📌 Question:

You receive a complaint that a user cannot access Instagram and YouTube on their laptop, but WhatsApp Web works fine. Write down the first 5 troubleshooting steps you would perform to identify if the issue is with hardware, software, or network configuration.

✅ Answer:

Since **WhatsApp Web is working**, the laptop already has some internet connectivity. Therefore, I would check the problem step-by-step.

🔍 Step 1: Check the Internet Connection

First, check whether the laptop is properly connected to WiFi.

Open **Command Prompt** and run:

```
ping 8.8.8.8
```

- ✅ If the ping works, basic internet connectivity is available.
 - ❌ If it fails, investigate the WiFi, router, cable, or network connection.
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🌐 Step 2: Test Other Websites

Open other websites such as Google.

- ✅ If other websites work, the problem may be specific to Instagram and YouTube.
 - ❌ If many websites fail, investigate the network connection or DNS.
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🌍 Step 3: Try Another Browser

Open Instagram and YouTube using another browser such as:

- Google Chrome
- Microsoft Edge
- Firefox

If the websites work in another browser, the problem may be related to the original browser.

Step 4: Clear Browser Cache and DNS Cache

First, clear the browser cache.

Then open **Command Prompt as required** and run:

```
ipconfig /flushdns
```

This removes old DNS information stored on the laptop and allows DNS information to be requested again.

Step 5: Check IP and DNS Configuration

Run:

```
ipconfig /all
```

Check the following:

- **IPv4 Address**
- **Default Gateway**
- **DNS Servers**
- Network Adapter status

If the DNS configuration is incorrect, some websites may not open even though other internet services are working.

Troubleshooting Summary

Step	What to Check	Purpose
1	Internet + Ping	Check basic connectivity
2	Other Websites	Identify website-specific problem
3	Another Browser	Check browser/software issue
4	Cache + DNS	Remove possible old DNS/cache data
5	IP + DNS Settings	Check network configuration

Conclusion:

These five steps help determine whether the problem is caused by **hardware, browser/software, DNS, or network configuration**. Troubleshooting should start with simple checks and then move toward more detailed network configuration.



FINAL CONCLUSION

Network troubleshooting is an important skill for every network administrator. A problem should be checked **systematically instead of changing many settings at once**.

★ Basic Troubleshooting Method:



Check Hardware



Check Connections



Check Device Settings



Test Network Connectivity



Check Software/DNS



Apply the Fix



Test Again

Following this method helps identify network problems faster and reduces unnecessary changes to the network.



Key Learning

“Check the simplest possible cause first, test the connection, identify the fault, fix it, and test again.”